Project Executable Files

Date	14 July 2026
Team ID	-
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package. json / dependency file	Yes
4	Database schema / seed files (if applicable)	Yes
5	Environment configuration file (.env.example similar)	Yes
6	Deployed application URL (if hosted)	no
7	APK / Executable binary (if applicable for mobile/desktop apps)	n/a

8	Dockerfile / Containerization config (if applicable)	No
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

Voice-Based Concept Understanding Analyser (VBCUA)

```
VBCUA/
│
├── app.py
├── requirements.txt
├── README.md
├── .env.example
│
├── assets/
│   ├── images/
│   ├── icons/
│   └── styles/
│
├── audio/
│   ├── uploads/
│   ├── processed/
│   └── waveforms/
│
├── database/
```

```
│   ├── database.db
│   ├── db.py
│   └── queries.py
│
├── models/
│   ├── whisper_model.py
│   ├── semantic_model.py
│   ├── audio_analysis.py
│   ├── keyword_match.py
│   ├── scoring.py
│   └── feedback.py
│
├── pages/
│   ├── Home.py
│   ├── About.py
│   ├── Analyze.py
│   ├── Dashboard.py
│   └── Reports.py
│
├── reference_concepts/
│
├── reports/
│
├── utils/
│   ├── helper.py
│   ├── visualization.py
│   ├── pdf_generator.py
│   └── reference_loader.py
│
└── tests/
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not deployed (Runs locally using Streamlit)
Login Credentials (Demo)	No login required
Platform / Hosting Provider	Local System (Streamlit)
Repository Link	SK-P-BibiAyesha/skill-wallet-google-cloud-generative-ai: Voice-Based Concept Understanding Analyser using Faster Whisper and Sentence-BERT
Demo Video Link	

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

1. Install Python 3.10 or above.
2. Clone the GitHub repository.
3. Open the project folder in Visual Studio Code.
4. Create and activate a virtual environment.
5. Install all required packages using:

   pip install -r requirements.txt

6. Ensure Faster Whisper and other dependencies are installed.
7. Run the application using:

   streamlit run app.py

8. Open the displayed localhost URL in your browser.
9. Upload an audio file, select a topic, and click "Analyze Audio" to view the results and download the PDF report.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Processing long audio files may take additional time	Expected due to AI transcription
2	Internet connection is required during initial model download	Download occurs only once
3	Supports only supported audio formats (WAV, MP3, M4A, OGG, WEBM)	Convert unsupported files before uploading